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LATE PALEOZOIC REMAGNETIZATION OF THE TRENTON LIMESTONE

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Abstract. Samples of the Middle Ordovician Trenton Limestone were collected from 8 sites in Quebec, Ontario, and New York State for paleomagnetic study. Stepwise thermal demagnetization reveals a very well defined characteristic magnetization that has a corresponding paleomagnetic pole at 53.0° N, 126.9° E (K=97.7, A-95=2.7°). This pole is not at all close to other Ordovician poles from cratonic North America but is near Late Carboniferous and Early Permian ones, indicating that the Trenton was remagnetized in the Late Paleozoic. An earlier paleomagnetic investigation of the Trenton gave a pole that did not appear to be Late Paleozoic in age. However, we demonstrate that this earlier study failed to isolate the ancient characteristic magnetization.

Rock magnetic experiments and analyses of magnetic extracts show that the carrier of magnetization is Fe_3O_4 that does not contain Ti or other elements in solid solution that would indicate a detrital source. The magnetite and its associated remanence appear to have a diagenetic origin.

Introduction

Sedimentary carbonates of Trentonian age form one of the most widespread rock units in North America. The Trenton Limestone in New York State has been the object of at least two previous paleomagnetic investigations. Graham (1954, 1956) reported well-clustered natural remanent magnetization (NRM) directions that have a mean declination/inclination of 179°/+82°. McElhinny and Opdyke (1973) resampled the same section at Trenton Falls, New York that was sampled by Graham. They showed that Graham's results, which were obtained without any demagnetization, are contaminated with a component of remanence aligned with the present-day magnetic field. After alternating field (AF) demagnetization they found a mean direction of 173°/+18°. Further, they showed (as had Graham) that the magnetization was acquired after a syndepositional slump but that the minimum and maximum axes of magnetic susceptibility anisotropy show a typically sedimentary fabric. They concluded that the magnetization was due to early post-depositional reorientation of the magnetite carrier grains in the ambient Ordovician field.

Several factors prompted us to undertake yet another investigation of Trenton paleomagnetism. Recent studies of Paleozoic carbonates from eastern North America (eg. Scotese et al., 1982; McCabe et al., 1983; Wisniowiecki et al., 1983) have demonstrated that some carbonate rock units were chemically remagnetized during the late Paleozoic by the diagenetic development of new magnetite. The Trenton pole from McElhinny and Opdyke (1973) is based on reversed-only directions with typically Kiaman Interval declinations, and thus gives rise to suspicions that it may be the result of remagnetization during the Late Paleozoic. Cryogenic magnetometers have only recently come into routine use in paleomagnetism. Their unprecedented sensitivity allows the complete stepwise demagnetization of even weakly magnetic rocktypes such as limestones. This study utilizes recent advances in instrumentation and methodology in paleomagnetism to test whether the Trenton magnetization was acquired during the Ordovician or at some later time.

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Sampling and Laboratory Procedures

Oriented hand samples of Trenton Limestone were collected from flat-lying strata at 8 sites. The site localities cover a broad area in Quebec, Ontario, and New York State (Fig. 1). Sites 7 and 8 are from the same Trenton Falls section sampled by Graham and by McElhinny and Opdyke. In the laboratory, samples were cored and standard paleomagnetic specimens prepared. Usually 2-3 specimens were cut from a sample, though in some cases shale partings made it impossible to obtain good specimens. Some of the samples were set aside for rock magnetic or dissolution experiments. Specimens were demagnetized in the laboratory using both AF and thermal treatments and measured with an ScT cryogenic magnetometer. Each specimen was demagnetized in stepwise fashion. The demagnetization data were plotted on orthogonal demagnetization diagrams and the directions of the component magnetizations were determined by vector subtraction (Zijderveld, 1967).

To obtain extracts of the magnetic minerals for direct observation, representative Trenton samples were powdered in an alumina-ceramic grinding dish, and the carbonate was dissolved with buffered acetic acid (1.5 M; pH=4.0). The magnetic fraction of the insoluble residue was extracted from a water suspension with a home-built magnetic extractor. Magnetic extracts were studied using scanning electron microscopy energy dispersive X-ray analysis and powder X-ray diffraction techniques.

Results

Demagnetization studies demonstrate that the characteristic magnetizations were not satisfactorily isolated by AF demagnetization, but that they were using thermal treatments. This is illustrated by two representative demagnetization diagrams in Fig. 2, one showing the behavior of the magnetization during AF and the other thermal demagnetization. The rather curved trajectory of the endpoints of the magnetization vectors during AF demagnetization suggests that the (viscous?) magnetization aligned with the present field has a broad coercivity spectrum that overlaps that of the ancient (characteristic) component of remanence. However, blocking temperature spectra appear not to overlap in the diagram showing behavior during thermal demagnetization. In the latter case there appears to be good separation of the blocking temperature spectra of the present-day and ancient magnetizations. The overlapping coercivity spectra make the isolation of the characteristic magnetization

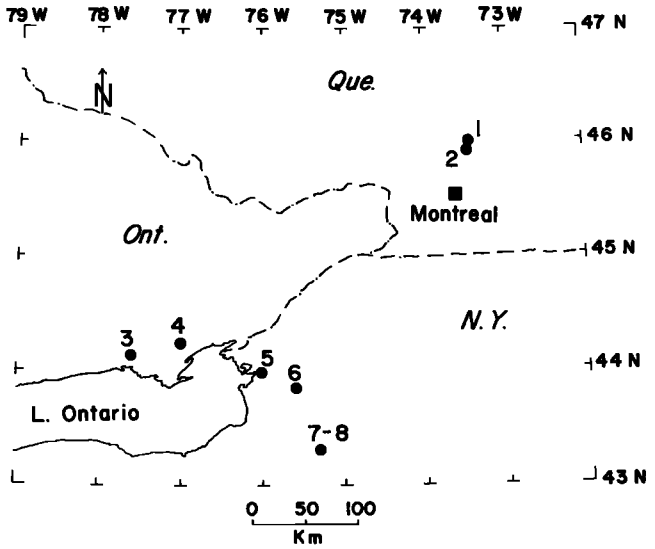


Fig. 1. Map of the study area showing the locations of the sampling sites.

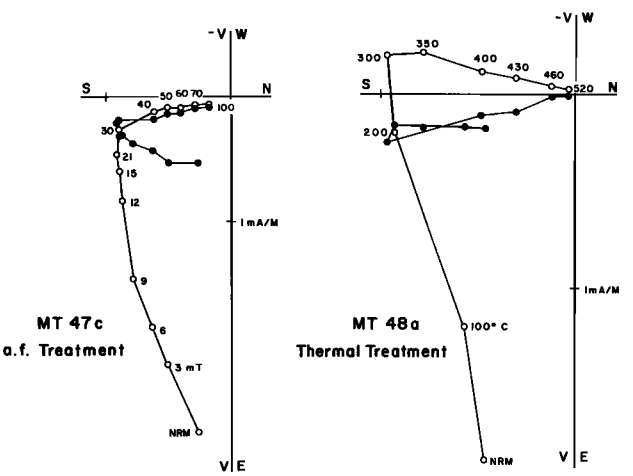


Fig. 2. Representative orthogonal demagnetization diagrams (Zijderveld, 1967). Open circles are the projections of the endpoints of the magnetization vectors onto a north-south vertical plane, closed circles are the projections onto the horizontal plane. Results of AF and thermal demagnetizations for two specimens from the same site at Trenton Falls, New York are shown for comparison.

impossible with AF treatment. Consequently, we used only thermal demagnetization data in the calculation of the paleomagnetic pole.

Sample mean directions are plotted in Fig. 3. Directions of the NRM before treatment cluster well with a steeply down mean similar to the result obtained by Graham (1954,1956). The directions of the recent viscous magnetization fall near the present field direction while characteristic directions are well clustered with a mean direction of 167°/-18°. Sample mean data for the characteristic magnetization are listed in Table 1 along with means and pertinent statistics. Data from two samples that did not yield interpretable characteristic magnetizations are omitted from the table.

The mean characteristic direction we obtained is plotted in Fig. 4 along with the mean directions from Graham (1954,1956) and from McElhinny and Opdyke (1973). The declination from each of these three studies is south-southeasterly but the inclinations differ markedly. Graham's result is steeply down, McElhinny and Opdyke's is shallow down (+18°) while the inclination we obtained is shallow up (-18°). All of these directions lie in approximately the same plane (represented as a great circle in Fig. 4), a plane that also includes the present field direction. The simplest explanation for this is that Trenton NRM is the resultant of two directions, one aligned with the present field, the other south-southeasterly and shallow up. The difference between the direction reported by McElhinny and Opdyke and that obtained in the present investigation is due to incomplete removal of the present-day component of remanence (because of its high coercivities) in the former study.

Rock magnetic studies have been undertaken to shed light on the physical properties of the carriers of magnetization. Acquisition of isothermal remanent magnetization (IRM) experiments reveal that representative specimens acquire IRM rapidly and reach saturation below 300 mT, implying that magnetite is the only carrier present. Other rock magnetic studies, acquisition of viscous magnetization and anisotropy of magnetic susceptibility are in progress.

Direct observation of the magnetic phases extracted from Trenton samples shows that only magnetite is present in significant quantities. Powder X-ray diffraction patterns obtained with a Gandolfi camera and Fe k-alpha radiation have only magnetite reflections. Under the scanning electron microscope the individual grains were found to be mostly irregular in shape and many appeared to have been freshly broken, probably when the sample was powdered. To test whether large grains are present that may have

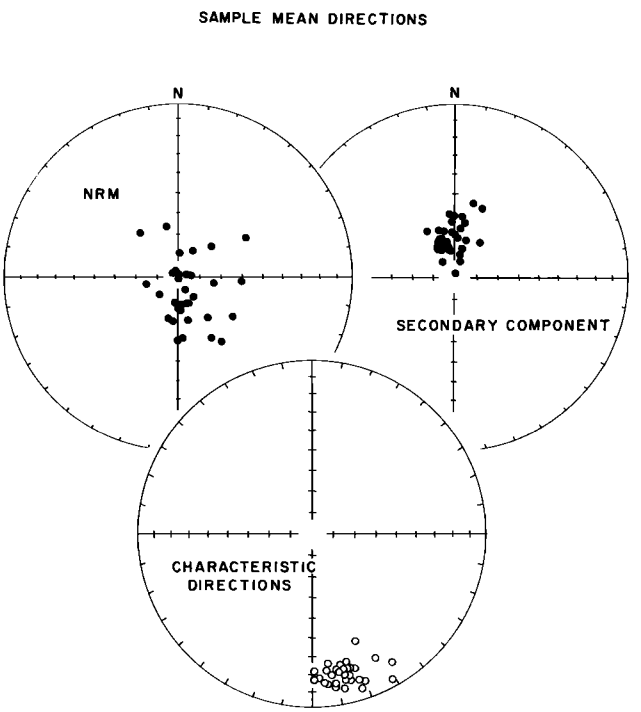


Fig. 3. Sample mean directions in equal area projection from all thermally demagnetized samples.

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been broken during extract preparation, a coarse crushed rock sample was dissolved in acid and extracted (a process that took several weeks). Though a broad range of grain sizes was present in this extract, magnetite grains as large as 3 mm were observed. Grain composition was studied using qualitative energy dispersive x-ray analysis. Analyses of individual grains revealed that about 95% of the grains are pure Fe_3O_4 without detectable Ti or other elements in solid solution. A study to characterize the extracts using scanning as well as transmission and analytical scanning-transmission electron microscopic techniques is still in progress.

Table 1. Summary of paleomagnetic results.

(Site) Sample	N/N_o	k	Direction Decl/Incl	Pole N lat, E long
(1)				
MT 1	1/1		168/-24	55.0, 127.3
MT 2	2/2	55.3	165/-17	50.5, 130.3
MT 4	2/2	116.7	164/-21	52.3, 132.9
MT 5	2/2	193.9	162/-20	51.2, 135.6
MT 6	1/1		162/-13	47.6, 133.7
(2)				
MT 7	1/1		174/-22	55.1, 116.9
MT 8	2/2	711.0	167/-18	51.7, 122.6
MT 11	2/2	1049.9	153/-21	48.1, 148.5
MT 12	2/2	93.0	167/-21	53.3, 128.3
(3)				
MT 14	2/2	22.6	165/-14	50.7, 126.3
MT 15	1/1		165/-21	54.2 128.2
MT 16	2/2	7.9	151/ -5	41.1, 142.4
MT 17	1/1		148/-14	43.2, 148.6
MT 18	2/2	17.4	179/-17	54.5, 104.1
(4)				
MT 19	2/2	56.7	179/-22	57.1, 104.8
(5)				
MT 25	1/1		158/-34	58.6, 147.0
MT 27	1/1		170/-22	56.3, 121.9
(6)				
MT 31	1/1		172/-19	55.4, 118.4
MT 33	1/2		160/-11	48.0, 135.0
MT 34	2/2	158.6	171/-12	51.5, 118.9
MT 35	2/2	48.1	165/-25	56.8, 131.8
MT 36	2/2	83.3	162/ -7	46.8, 131.2
(7)				
MT 37	2/2	6749.7	174/-14	53.4, 114.9
MT 38	2/2	264.3	175/-15	54.1, 113.4
MT 39	2/2	1600.0	173/-26	59.8, 118.5
MT 41	3/3	541.6	177/-18	55.8, 110.2
MT 42	1/1		169/-20	55.7, 124.3
(8)				
MT 43	2/2	1052.9	171/-14	53.0, 123.8
MT 46	2/2	329.4	168/-10	50.3, 128.3
MT 48	2/2	13131.3	167/-15	52.6, 126.4
Means (n=30):			166.7/-17.9	53.0, 126.9
			k=73.9	K=97.7
			$\alpha_{95}=3.1^o$	A-95=2.7°

N is the number of specimen characteristic directions used in computing the sample mean and N_o is the number of specimens thermally demagnetized. K, k, A-95 and α_{95} are statistical parameters (Fisher, 1953).

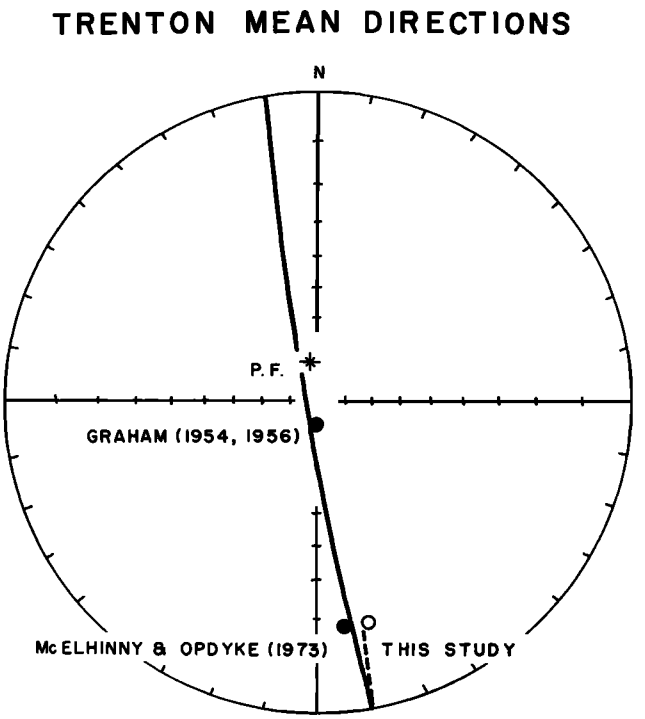


Fig. 4. Mean formation directions in equal angle projection from three paleomagnetic studies of the Trenton Limestone as indicated. P. F. is the present field direction. The great circle represents a nearly vertical plane that approximately includes all of these directions.

Discussion

A very well defined characteristic magnetization has been isolated in 30 Trenton Limestone samples (52 specimens) using thermal demagnetization. The paleomagnetic pole is located at 53.0° N, 126.9° E (K=97.7; A-95=2.7°). This pole is plotted in Fig. 5 along with the pole found by McElhinny and Opdyke (1973). It is clear from the figure that the pole we have obtained is near Late Carboniferous and Early Permian poles for North America but is not at all close to other Ordovician poles, indicating remagnetization in the Late Paleozoic. This new Trenton pole is close to the Onondaga Limestone pole (Kent, 1979) which failed the fold test. It is also virtually coincident with the remagnetized Helderberg Limestone pole reported by Scotese et al. (1982). A fold test on the Helderberg magnetization (McCabe et al., 1983) demonstrated a syn-Alleghenian acquisition of the characteristic remanence and proved its secondary nature. The Trenton thus appears also to carry a Late Paleozoic secondary magnetization. Since the Trenton magnetization reported by McElhinny and Opdyke (1973) is the resultant of two components, a Late Paleozoic one and an incompletely removed recent one, that result should no longer be considered as representative of the Ordovician field for North America.

The Trenton Late Paleozoic remagnetization cannot be of thermal origin since paleotemperature estimates indicate only moderate burial temperatures in the area sampled (Epstein et al., 1977). The end-member purity of most of the magnetite observed in Trenton magnetic extracts suggests that it is not of high temperature (detrital) origin. While a biogenic

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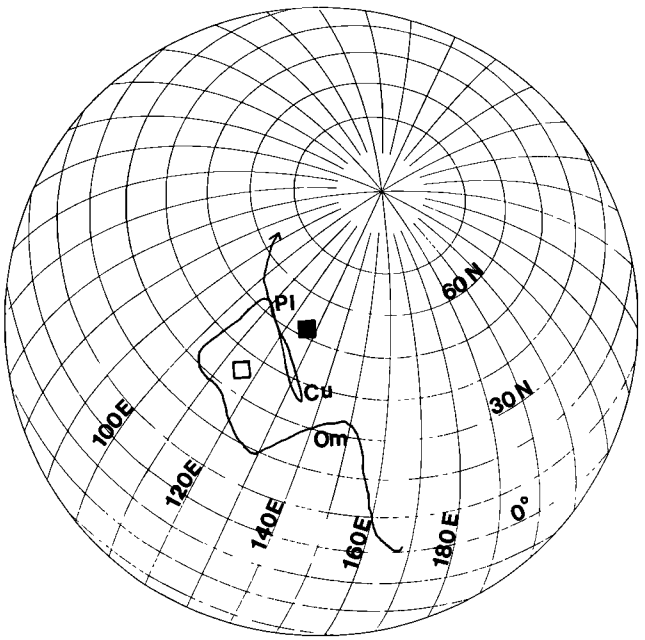


Fig. 5. Orthographic projection showing the locations of the paleomagnetic poles obtained in this study (closed symbol) and the one reported by McElhinny and Opdyke, 1973 (open symbol). Curve is the Paleozoic portion of the apparent polar wander path for North America after Van der Voo (1981).

source for the magnetite cannot be discounted on presently available information, the evidence from magnetic extract analyses are permissive of a diagenetic origin. In light of the secondary nature of the characteristic remanence, a diagenetic origin for both the magnetite and the magnetization it carries seems to be the most likely explanation. An unsolved problem is the "sedimentary" magnetic fabric observed by Graham in Trenton samples. A study to shed light on the origin of the observed anistropy of magnetic susceptibility is in progress.

Chemical remanent magnetizations in limestones and dolomites carried in diagenetic magnetite have now been identified over a large area in North America. Such magnetizations, once thought to be quite rare, now appear to be rather common in older sedimentary carbonates. Though we do not understand the ultimate cause of these magnetizations, we currently favor a model in which Alleghenian mountain building events caused deep basinal fluids to migrate updip towards the craton, and that these fluids were the agents responsible for chemical changes resulting in the observed remagnetizations.

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